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Sec: Star SC

S-Block Elements

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SINGLE ANSWER TYPE

- Which of the following is known as fusion mixture?
A) $\text{Na}_2\text{CO}_3 + \text{K}_2\text{CO}_3$ B) $\text{Na}_2\text{CO}_3 + \text{KOH}$ C) $\text{K}_2\text{CO}_3 + \text{KHCO}_3$ D) $\text{NaOH} + \text{KOH}$
- Which of the following is used in the treatment of manic depression?
A) Li_2CO_3 B) Na_2CO_3 C) CaCO_3 D) K_2CO_3
- Which of the following can be used in space capsules?
A) Na_2O B) KO_2 C) Na_2CO_3 D) All of these
- Which of the following statements is correct for? CsBr_3 ?
A) It is a covalent compound B) It contains Cs^{3+} and Br^- ions
C) It contains Cs^+ and Br_3^- ions D) It contains Cs^+ , Br^- and lattice Br_2 molecule
- Sodium forms Na^+ and not Na^{2+} because:
A) Sodium contains only one electron in outer most shell
B) First ionization potential is small and the difference in first and second ionization Potentials is large (Na^+ has neon configuration where as Na doesn't)
C) Radius of Na^{2+} is much smaller than of Na^+ D) It has Argon configuration
- Which of the following is lightest metal?
A) Na B) K C) Cs D) Rb
- In vapour the order of reducing property of IA group elements is
A) $\text{Li} > \text{Na} > \text{K} > \text{Rb}$ B) $\text{Li} < \text{Na} < \text{K} < \text{Rb}$ C) $\text{Na} > \text{Li} > \text{K} > \text{Rb}$ D) $\text{Na} > \text{K} > \text{Li} > \text{Rb}$
- Sodium can be extracted from.
A) Fused caustic soda B) Aqueous sodium chloride solution
C) Brine solution D) Potash lye
- Hybridization of carbon in carbonate ion is
A) sp^3 B) dsp^2 C) sp^2 D) sp

MULTIPLE ANSWER TYPE

- Mg burns in air to give
A) MgO B) Mg_3N_2 C) MgCO_3 D) $\text{Mg}(\text{OH})_2$
- The alkali metals dissolve in NH_3 it is found that
A) The dilute solutions are blue but colour changes to bronze with increasing concentration
B) The blue colour is due to the presence of solvated electrons.
C) The blue solutions are paramagnetic
D) In concentrated solutions the ammoniated ions are bound by the free unpaired electrons Called an expanded metal
- Select correct statement (s)
A) Beryllium and magnesium hydride are covalent and polymeric
B) CaH_2 ; SrH_2 and BaH_2 are ionic C) BeH_2 contains three-centre two electron bond.
D) Ionic hydrides react with water or acid forming hydrogen.

13. Sodium sulphate is soluble in water but barium sulphate is sparingly soluble
 A) The hydration energy of Na_2SO_4 is more than its lattice energy.
 B) The lattice energy of barium sulphate is more than its hydration energy.
 C) The lattice energy does not play role in solubility.
 D) The lattice energy of sodium sulphate is more than its hydration energy..
14. Flame test is given by
 A) Ba B) Ca C) Sr D) Be
15. Which of the following elements shows diagonal relationship
 A) Li and Mg B) B and Si C) C and Al D) Be and Mg

PASSAGE-1 Read the following passage and answer the questions:

The solubility of most of the salts of group 1 elements in water decreases on descending the group. For a substance to dissolve, the energy evolved when the ions are hydrated is more than the energy required to break the crystal lattice. If the solid is insoluble, the hydration energy is less than the lattice energy. Lattice energy is an enthalpy ΔH° term and we should use ΔG° the standard free energy for converting the crystalline salt into gaseous ions. The reason why the solubility of most group metals decreases on descending the group is that the lattice energy only changes slightly but the free energy of hydration changes rather more

16. Which of the following shows highest of hydration energy?
 A) Rb^+ B) K^+ C) Na^+ D) Li^+
17. Which of the following shows highest lattice energy?
 A) LiF B) $LiCl$ C) RbF D) RbI
18. Which of the following has least solubility in aqueous solution
 A) NaF B) KF C) LiF D) all are insoluble

INTEGER ANSWER TYPE

19. What is the number of penultimate electron present in lithium ion?
 20. How many elements in the periodic table are lack of neutron (i.e without neutron)?
 21. What is the number of hydrogen atoms present in hydrolith?
 22. Among the following species, how many species are covalent? $BeCl_2, NaCl$ & MgF_2

23. **Column-I**

- A) Na
 B) K
 C) Be
 D) Li

Column-II

- P) Most -ve reduction potential value
 Q) No flame coloration of the chloride
 R) Thermally stable carbonate
 S) Peroxide.

24. **Column-I**

- A) KO_2
 B) H_2O_2
 C) D_2O
 D) H_2

Column-II

- P) Used in nuclear reactor as moderator
 Q) One of the component in water gas
 R) act as both oxidizing and reducing agent
 S) used in submarines for purification of air (i.e absorbs CO_2 from air and release O_2 into air)

25. **Column-I**

- A) Sylvine
 B) Rock salt
 C) Soda lime
 D) Pearl's ash

Column-II

- P) KCl
 Q) $NaCl$
 R) $NaOH + CaO$
 S) K_2CO_3

KEY

1	A	2	A	3	B	4	C	5	B	6	B	7	B	8	A
9	C	10	AB	11	ABCD	12	ABCD	13	AB	14	ABC	15	AB	16	D
17	A	18	C	19	0	20	1	21	2	22	1				
23	A-RS,B-RS,C-Q,D-PS					24	A-S,B-R,C-P,D-Q				25	A-P,B-Q,C-R,D-S			

SINGLE ANSWER TYPE

- The compound X on heating gives a colourless gas. The residue is dissolved in water to obtain Y. Excess CO_2 is bubbled through aqueous solution of Y, Z is formed. Z on gentle heating gives back X. The compound X is
 A) CaCO_3 B) Na_2CO_3 C) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ D) K_2CO_3
- The fluoride which is most soluble in water is:
 A) CaF_2 B) BaF_2 C) SrF_2 D) BeF_2
- When $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ is strongly heated, then it forms :
 (A) MgO (B) $\text{Mg}(\text{OH})_2$ (C) $\text{Mg}(\text{OH})\text{Cl}$ (D) MgCl_2
- In polymeric $(\text{BeCl}_2)_n$, there are:
 (A) three centre two-electron bonds (B) three centre three-electron bonds
 (C) two centre three-electron bonds (D) two centre two-electron bonds.
- An element A is invariably divalent and its oxide is soluble in excess of NaOH and the cation of A does not (with stable e configuration) have electrons in d-subshell.
 (A) Ca (B) Be (C) Zn (D) Pb
- Choose the incorrect statement
 (A) Reducing power in aqueous solution is maximum for Li metal
 (B) Electron affinity order $\text{O}^+ > \text{O} > \text{O}_2^{-2} > \text{O}^{-2}$
 (C) order of oxidation number of oxygen $\text{O}_3 > \text{KO}_2 > \text{BaO}_2 > \text{K}_2\text{O}$
 (D) pH of aqueous solution $\text{LiCl} > \text{BeCl}_2 > \text{MgCl}_2 > \text{AlCl}_3$
- The compound 'X' on heating gives a colour less gas, the residue is dissolved in water to obtain 'Y'. Excess CO_2 is passed through aq solution of Y when Z is formed . Z on gentle heating gives back X, the compound X is
 (A) Na_2CO_3 (B) CaCO_3 (C) $\text{Ca}(\text{HCO}_3)_2$ (D) NaHCO_3
- S-I : Beryllium salts are acidic when dissolved in pure water
 S-II : Be-O bond is very strong thus it weakens the O-H bond in hydrated ion
- Which of the following trends are true (T) (or) false (F) ?
 Solubilities
 (1) $\text{Cs}_2\text{SO}_4 > \text{Rb}_2\text{SO}_4 > \text{Li}_2\text{SO}_4$ (2) $\text{BeSO}_4 < \text{CaSO}_4 < \text{BaSO}_4$ (3) $\text{Mg}(\text{OH})_2 > \text{Ca}(\text{OH})_2 > \text{Sr}(\text{OH})_2$
 (A) TFT (B) TFF (C) FFF (D) TFT
- The element with which sodium hydroxide reacts is
 (A) Carbon (B) Nitrogen (C) Phosphorus (D) Hydrogen
- Among the s-block elements, the metals that do not react with cold water is/ are
 (A) Li, Mg, Be (B) Li, Be (C) Mg, Be (D) Be only
- Which of the following reagents have a different reaction with Be as compared to other Group 2 metals?
 (A) dil HCl (B) CO_2 gas (C) dil HNO_3 (D) NaOH
- Select the correct statement
 (A) CaC_2 have two σ and one π bond
 (B) In $(\text{SiH}_3)_3\text{N}$ lone pair present on N atom does not undergo back bonding
 (C) Lowest Unoccupied Molecular Orbital level for C_2 molecule is $\sigma 2p_z$ orbital
 (D) XeF_2 has V-shape with three lone pairs.
- Pick out statement(s) which is(are) not true about diagonal relationship of Li and Mg.

